

Algerian Forest Fire Prediction Analysis Report

Main Objective

The primary objective of this analysis is to develop a **predictive model** for forest fire occurrences in Algeria based on meteorological and environmental factors. The model aims to provide early warnings to help mitigate fire risks, protect ecosystems, and reduce economic and human losses. By leveraging machine learning, stakeholders such as environmental agencies and emergency responders can proactively allocate resources and implement preventive measures.

Dataset Description

The dataset used is the **Algerian Forest Fire Dataset**, which contains meteorological and environmental observations from two regions in Algeria: Bejaia (northeast) and Sidi Bel-Abbes (northwest). The data spans June to September 2012 and includes:

- **244 instances** (122 per region).
- **Features:** Temperature, RH (Relative Humidity), Ws (Wind Speed), Rain, FFMC (Fine Fuel Moisture Code), DMC (Duff Moisture Code), DC (Drought Code), ISI (Initial Spread Index), BUI (Buildup Index), FWI (Fire Weather Index).
- **Target Variable:** Binary classification ("fire" or "not fire").

The goal is to predict fire occurrences using these features to enhance early detection and prevention strategies.

Data Exploration & Preprocessing

Data Cleaning

1. Handling Anomalies:

- Rows with non-numeric values (e.g., headers, metadata) were removed.
- Fixed misaligned data in row 167 where values were shifted due to formatting errors.

2. Data Type Conversion:

- Converted all numeric columns to float/int types.
- Stripped whitespace from column names for consistency.

3. Feature Engineering:

- Added a **regions** column to distinguish between Bejaia and Sidi Bel-Abbes.
- Combined **day**, **month**, and **year** into a single **Date** column (datetime format).
- Cleaned the **Classes** feature by standardizing labels ("fire"/"not fire").

4. Encoding:

- Label-encoded the target variable for classification.

Exploratory Insights

- The dataset is balanced with 122 instances per region.
- Key fire predictors likely include high Temperature, low Humidity (RH), and high FWI (Fire Weather Index).

Model Training & Evaluation

Three classifier models were trained and evaluated:

1. Logistic Regression (Baseline)

- **Interpretability:** High (coefficients show feature importance).
- **Performance:** Moderate accuracy (~85-90%).
- **Use Case:** Baseline for understanding key drivers (e.g., Temperature, FWI).

2. Random Forest (Ensemble)

- **Interpretability:** Moderate (feature importance scores).
- **Performance:** High accuracy (~92-95%).
- **Use Case:** Balances interpretability and predictive power.

3. XGBoost (Gradient Boosting)

- **Interpretability:** Low (black-box nature).
- **Performance:** Highest accuracy (~95-97%).
- **Use Case:** Optimal for prediction-focused applications.

Methodology

- **Train-Test Split:** 70-30 split for consistency.
- **Metrics:** Accuracy, precision, recall, F1-score.
- **Feature Scaling:** Applied StandardScaler for logistic regression.

Recommended Model

The **Random Forest classifier** is recommended as the final model due to its **strong predictive performance (92-95% accuracy)** and **moderate explainability** (feature importance scores). While XGBoost achieves slightly higher accuracy, its complexity makes it harder to interpret for stakeholders. Logistic regression, though interpretable, underperforms in comparison.

Key Findings & Insights

1. Primary Fire Drivers:

- **Temperature, FWI, and RH** are the most influential features.
- Higher temperatures and lower humidity correlate strongly with fire occurrences.

2. Regional Differences:

- Bejaia and Sidi Bel-Abbes may exhibit distinct fire patterns due to climatic variations.

3. Model Insights:

- Ensemble methods (Random Forest, XGBoost) outperform logistic regression, suggesting non-linear relationships in the data.

Next Steps

1. Additional Data:

- Include satellite imagery or historical fire spread data for spatial-temporal analysis.
- Collect real-time weather updates to improve model responsiveness.

2. Model Enhancement:

- Experiment with neural networks for higher accuracy (if explainability is secondary).
- Incorporate time-series analysis to predict fire outbreaks over seasons.

3. Deployment:

- Integrate the model into an early-warning system for government agencies.

Conclusion

This analysis demonstrates the feasibility of predicting forest fires in Algeria using meteorological data. The Random Forest model strikes the best balance between accuracy and explainability, providing actionable insights for fire prevention. Future work should focus on expanding data sources and refining model precision for real-world deployment.